(b) $\vec{F}_{\text{perp}} = \vec{0}$
(c) $W = -50$

49 (a) $\vec{F}_{\text{parallel}} = 3.846\vec{i} - 0.769\vec{j}$
(b) $\vec{F}_{\text{perp}} = -3.846\vec{i} - 19.231\vec{j}$
(c) $W = 20$

51 (a) $\vec{F}_{\text{parallel}} = \vec{0}$
(b) $\vec{F}_{\text{perp}} = \vec{F}$
(c) $W = 0$

53 No

55 (a) 2.24 m/sec
(b) 1.89 m/sec

57 $10/\sqrt{30}$

59 Averages; weightings
85.65%; class average

71 Can't take dot product of a scalar and a vector

73 Normal vector is $2\vec{i} + 3\vec{j} - \vec{k}$

75 $f(x, y) = (-1/3)x + (-2/3)y$

77 True

79 False

81 True

83 False

85 True

Section 13.4

1 $-\vec{i}$

3 $-\vec{i} + \vec{j} + \vec{k}$

5 $\vec{i} + 3\vec{j} + 7\vec{k}$

7 $7\vec{i} + \vec{j} + 4\vec{k}$

9 $-2\vec{k}$

11 $\vec{i} - \vec{j}$

13 $\vec{v} \times \vec{w} = -6\vec{i} + 7\vec{j} + 8\vec{k}$
$\vec{w} \times \vec{v} = 6\vec{i} - 7\vec{j} - 8\vec{k}$
$\vec{v} \times \vec{w} = -(\vec{w} \times \vec{v})$

15 $x - y - z = -3$

17 4

19 0

21 $4x + 26y + 14z = 0$

23 $3x - y - 2z = 0$

25 $4\vec{i} + 26\vec{j} + 14\vec{k}$

27 $4(x - 4) + 26(y - 5) + 14(z - 6) = 0$

29 (a) $\vec{u}$ and $-\vec{u}$ where
$\vec{u} = \frac{12}{13}\vec{i} - \frac{4}{13}\vec{j} - \frac{3}{13}\vec{k}$
(b) $\theta \approx 49.76°$
(c) $13/2$
(d) $13/\sqrt{29}$

31 Parallel to the z-axis

33 $\sqrt{38}/3$

35 (a) Increases force
(b) Ball moves down and to the left

37 (a) $\overrightarrow{P_1P_2} = \overrightarrow{P_3P_4} = 2\vec{i} + 4\vec{j} + 2\vec{k}$;
$\overrightarrow{P_1P_3} = \overrightarrow{P_2P_4} = 3\vec{i}$
(b) $\sqrt{180}$

51 $4\pi\vec{i}$

53 (a) $((u_2v_3 - u_3v_2)^2 + (u_3v_1 - u_1v_3)^2 + (u_1v_2 - u_2v_1)^2)^{1/2}$
(b) $|u_1v_2 - u_2v_1|$
(c) $m = (u_2v_3 - u_3v_2)/(u_2v_1 - u_1v_2)$,
$n = (u_3v_1 - u_1v_3)/(u_2v_1 - u_1v_2)$

55 Parallel, not perpendicular

57 $\vec{v} = (8\vec{i} - 6\vec{j})/5$

59 False

61 True

63 True

65 True

67 False

Chapter 13 Review

1 Scalar; -1

3 -1

5 $\vec{a} = -2\vec{j}$, $\vec{b} = 3\vec{i}$, $\vec{c} = \vec{i} + \vec{j}$,
$\vec{d} = 2\vec{j}$, $\vec{e} = \vec{i} - 2\vec{j}$, $\vec{f} = -3\vec{i} - \vec{j}$

7 $5\vec{i} + 30\vec{j}$

9 $3\sqrt{2}$

11 $3\vec{i} + 7\vec{j} - 4\vec{k}$

13 -3

15 $\vec{0}$

17 0

19 $\vec{0}$

21 $-5\vec{i} + 3\vec{j} + \vec{k}$ (multiples of)

23 (a) 4
(b) $-4\vec{i} - 11\vec{j} - 17\vec{k}$
(c) $3.64\vec{i} + 2.43\vec{j} - 2.43\vec{k}$
(d) $79.0°$
(e) 0.784.
(f) $2\vec{i} - 2\vec{j} + \vec{k}$ (many answers possible)
(g) $-4\vec{i} - 11\vec{j} - 17\vec{k}$.

25 $\pm(-\vec{i} + \vec{j} - 2\vec{k})/\sqrt{6}$

27 $\vec{n} = 4\vec{i} + 6\vec{k}$

29 $-3\vec{i} + 4\vec{j}$

31 $\vec{u}$ and $\vec{w}$; $\vec{v}$ and $\vec{q}$.

33 $\vec{F}_{\text{parallel}} = \vec{F}$
$\vec{F}_{\text{perp}} = \vec{0}$
$W = -10$

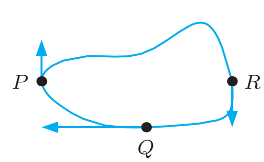
35 $\vec{F}_{\text{parallel}} = -(6/5)\vec{i} + (8/5)\vec{j}$
$\vec{F}_{\text{perp}} = (16/5)\vec{i} + (12/5)\vec{j}$
$W = -10$

37 $\vec{F}_{\text{parallel}} = 2\vec{j}$
$\vec{F}_{\text{perp}} = 5\vec{i}$
$W = 6$

39 (a) True
(b) False
(c) False
(d) True
(e) True
(f) False

41 548.6 km/hr

43



45 Parallel:
$3\vec{i} + \sqrt{3}\vec{j}$ and $\sqrt{3}\vec{i} + \vec{j}$
Perpendicular:
$\sqrt{3}\vec{i} + \vec{j}$ and $\vec{i} - \sqrt{3}\vec{j}$
$3\vec{i} + \sqrt{3}\vec{j}$ and $\vec{i} - \sqrt{3}\vec{j}$

47 2, 8

49 $2x - 3y + 7z = 19$

51 $\sqrt{6}/2$

53 (a) 1.5
(b) $y = 1$

55 $9x - 16y + 12z = 5$
0.23

57 $7.0710\vec{i} + 2.5882\vec{j} + 6.580\vec{k}$

59 228.43 newtons
$85.5°$ south of east

61 $|ywr - vzr + zus - xws + xvt - yut|$

63 (b) $(1, 1/\sqrt{3}, \sqrt{6}/6)$
(c) $109.471°$

Section 14.1

1 $f_x(3, 2) \approx -2/5$; $f_y(3, 2) \approx 3/5$

3 $-0.0493, -0.3660$
$-0.0501, -0.3629$

5 $\partial P/\partial t$:
dollars/month
Rate of change in payments with time
negative
$\partial P/\partial r$:
dollars/percentage point
Rate of change in payments with interest rate
positive

7 (a) Payment \$376.59/mo at 1% for 24 mos
(b) 4.7¢ extra/mo for \$1 increase
(c) Approx \$44.83 increase for 1% interest increase

9 (a) Negative
(b) Positive

11 $f_x > 0$, $f_y < 0$

13 $f_x < 0$, $f_y > 0$

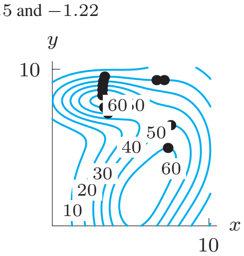
15 $f_T(5, 20) \approx 1.2°\text{F}/°\text{F}$

17 Positive, Negative, 10, 2, -4

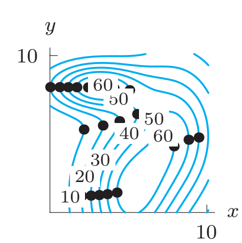
19 $\partial Q/\partial b < 0$
$\partial Q/\partial c > 0$

21 -1.5 and -1.22

23 (a)



(b)



25 (a) Negative
(b) Positive

27 (a) 2.5, 0.02
(b) 3.33, 0.02
(c) 3.33, 0.02

29 (i)(c); (ii)(a)

31 -2.5